Research Question

Can we use a stock's past returns and other technical indicators to predict whether its price will go up the next trading day?

Dataset

We will use the public Kaggle dataset Stock Market Signal: Predict Next-Day Returns, created by JSaleeby:

https://www.kaggle.com/competitions/stock-market-signal-predict-next-day-returns/data

The training data has about 440,000 rows from 100 anonymous stocks. Each row is one stock on one trading day. It has 27 features, such as past returns, moving averages, trading volume indicators, and volatility. The target is 1 if the stock price goes up the next day and 0 otherwise. We will predict whether the price increases, not by how much. We do not need to collect or scrape any other data.

The dataset page lists its license as CC BY-SA 4.0. This means we can use and share the data as long as we give proper credit. We will cite the creator, include links to the dataset and the license, explain any changes we make, and share any adapted data and figures under the same license. The data does not include personal information about anyone.

Methods and Course Connection

We plan to compare three models: logistic regression, LightGBM, and a small neural network called a multilayer perceptron (MLP). Logistic regression will also serve as our simple feature-based benchmark. LightGBM is a tree-based model that often works well on table-style data like ours. The MLP will have two or three hidden layers, so it can learn more complex patterns among past returns, volatility, and other indicators. We want to see if deep learning does better than the simpler models.

We will first check for missing values, remove the row ID, and standardize the numerical features using only the training data. We will use a validation set to pick a few model settings, such as the learning rate and the size of the hidden layers. For the MLP, we will use dropout and early stopping to reduce overfitting. All models will use the same data split so the comparison is fair. We will not touch the final test set until we have picked our final models.

This project connects to course topics such as classification, model selection, and measuring how well a model predicts. The MLP is where we use deep learning.

Baseline and Expected Results

Our baseline will give every stock-day the same predicted probability: the share of up-days in the training data. Since it gives every row the same score, its ROC-AUC is 0.5. We will set aside 20% of the labeled data for the final test and split the other 80% (development data) into training and validation sets to build and tune the models. We will mainly compare models using ROC-AUC, where 0.5 means the model is no better than random guessing. We will also report accuracy and log loss.

The data has been shuffled and has no dates, so we cannot split it into earlier and later periods. Our results will measure performance on randomly held-out observations, not performance on future data. Our final report will include a table comparing the models, ROC curves, and a discussion of which features seem useful. Even if the models do only about as well as the baseline, that still tells us how hard it is to predict next-day stock moves. Also, better prediction scores alone would not prove that a trading strategy makes money.

Group Members and Work Plan

Qidi An will prepare and explore the data. Jin Wang will build and tune the models. Yixun Lu will evaluate the results and create the figures. Everyone will help interpret the results and write the report. We plan to finish the project by December 11.